Developing the ambr250ht system as a scale-down model for perfusion cell culture

Zayla Schaeffer,[1] John Murphy[2]

[1] BioProcess Technologies and Engineering, BioPharmaceuticals Development, R&D, AstraZeneca, Gaithersburg, US

[2] Early Stage Formulation Sciences, BioPharmaceuticals Development, R&D, AstraZeneca, Gaithersburg, US

23 Sept 2024

Agenda

1 Motivation and perfusion background

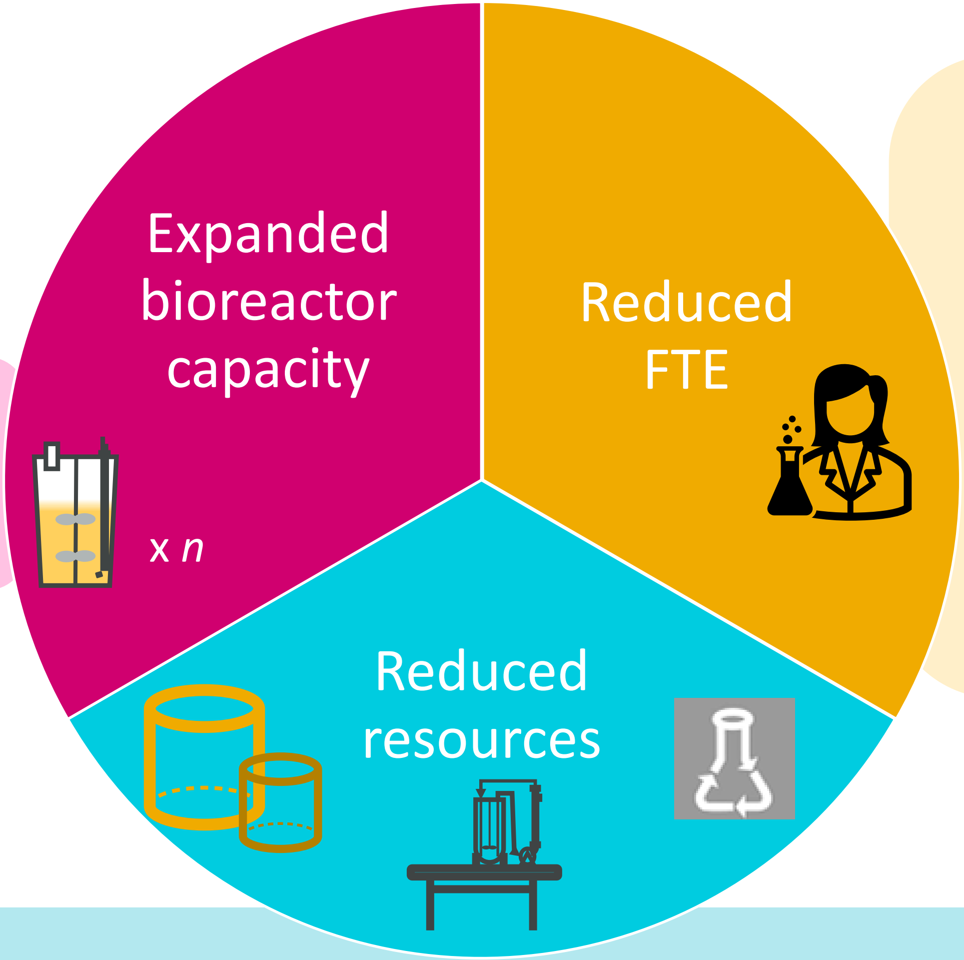
2 Improvements to the ambr250 platform

3 Scale-down model testing

4 Conclusions

The ambr250ht perfusion system offers reduced resources and expanded capacity over typical 3 L bench-scale systems

Run up to 24 bioreactors in the same bench space as 8 of 3 L perfusion bioreactors



In-run sampling time reduced by 50% due to automated bioreactor sampling to Nova Flex2 and other automation

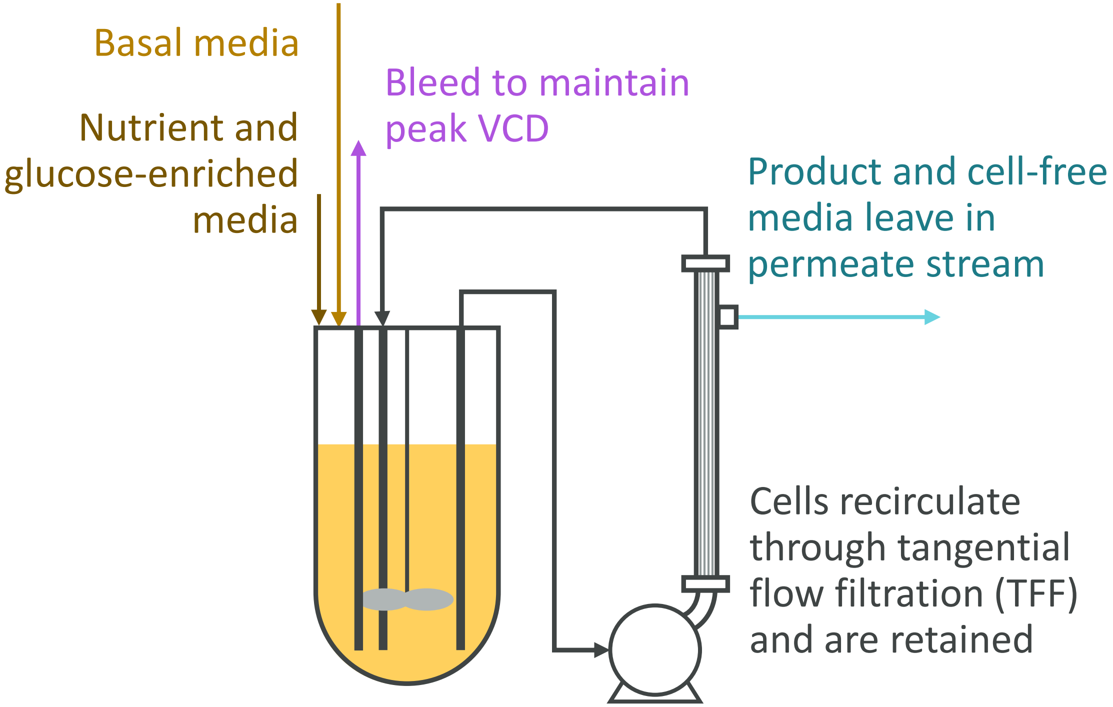
User tasks simplified to:

- sample handling
- reviewing software outputs
- replenishing on-deck consumables

Media usage decreased to 1/8 per bioreactor

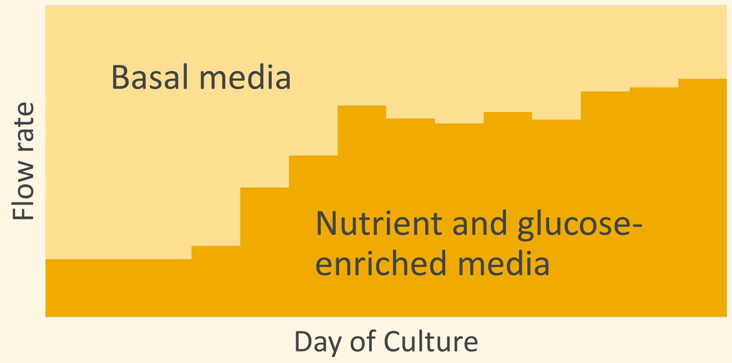
Reduced footprint for large experiments

Upstream cell culture perfusion platform grows cells to high densities

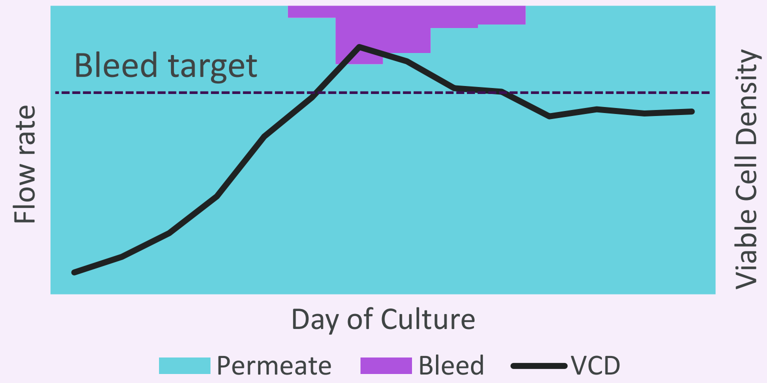


Example 3L perfusion bioreactor

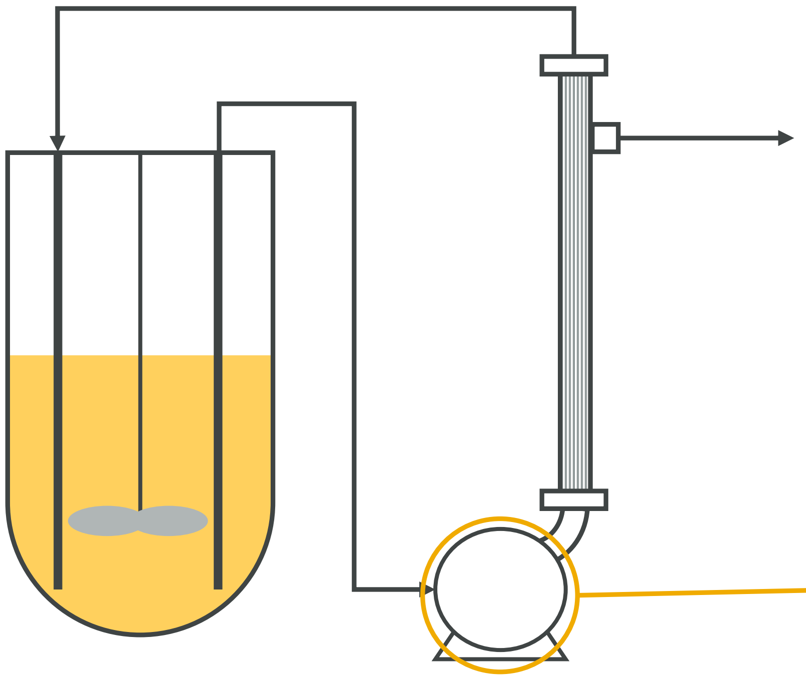
Feeding ratio of the two inlet feeds updated according to culture growth predictions and glucose measurements



Bleeding occurs after cell density bleed target is reached; **perfusion set point** (1-1.5 VVD) maintained by reducing permeate flow rate



Recirculation loop varies between scales



3-L+ scale

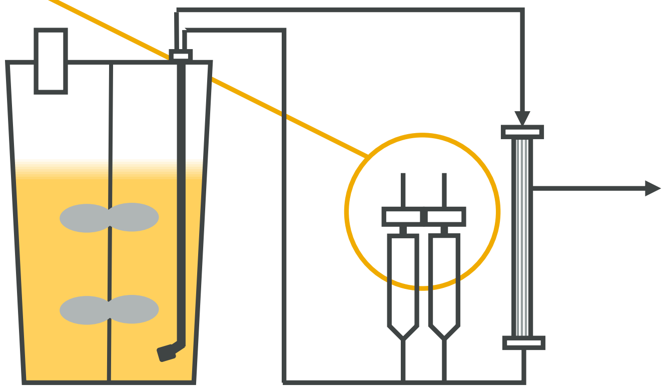
Centrifugal pump spinning at RPM range of 700-1100

RPM controlled by in-line flow meter reading

Air-enabled pumps: push and pull culture into each chamber

Not a significant source of shear stress

ambr250 scale



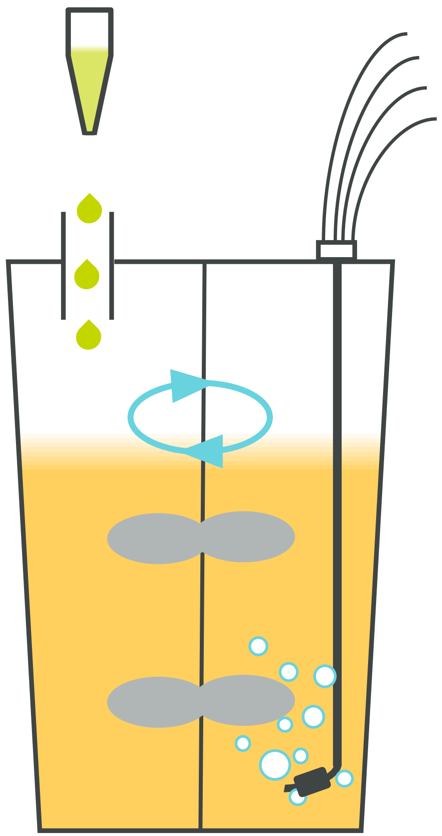
Additional scale-dependent differences

Physical scale:

- Sparger size, position, and resulting bubble size diameter
- Number of impellers, relative heights, and impeller power number

Control Strategies:

- Programming of available DO control loops
- 3-L uses level sensor or scale to control volume while ambr250 uses calibrated pumps and volume totalizer equations



Identifying and addressing ambr250 challenges

Early experiments identified challenges to using the ambr250ht perfusion as a scale-down model

User-driven antifoam control led to high FTE and frequent foam outs

- High level of consumable usage (300uL pipette tips) led to frequent replacement of tip boxes
- Scientist updating interval and volume of antifoam additions daily
- Regularly saw 10-20% of working volume lost

3-L bioreactors typically see no foam outs and require once-daily antifoam update

Working volume varied significantly from set point

- ambr250 volume controlled by pump calibrations and totalizers
- Identified up to 10-25% of original working volume missing by end of experiment with no root cause

3-L bioreactor working volume maintained by level sensor (scale also common)

Running separate seed and production bioreactors reduces total capacity of ambr250

- Alternatively, a manual drain-and-fill process is time consuming and can potentially introduce contamination

3-L bioreactors manually transition from seed to production

Scientists spent 2 hours* daily to keep experiment running

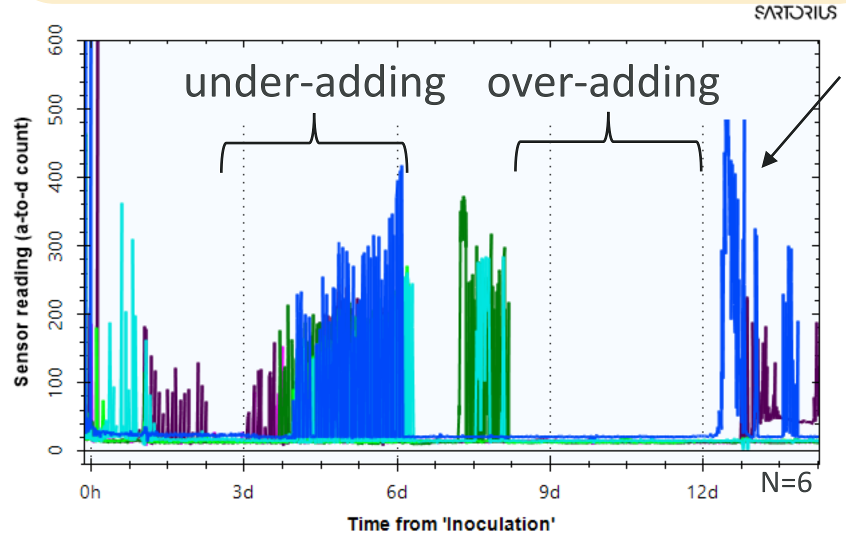
* For a 12-bioreactor system

Automated foam control strategy substantially reduces foam outs

Previous foam control strategy

Scientist updates frequency and volume of antifoam additions

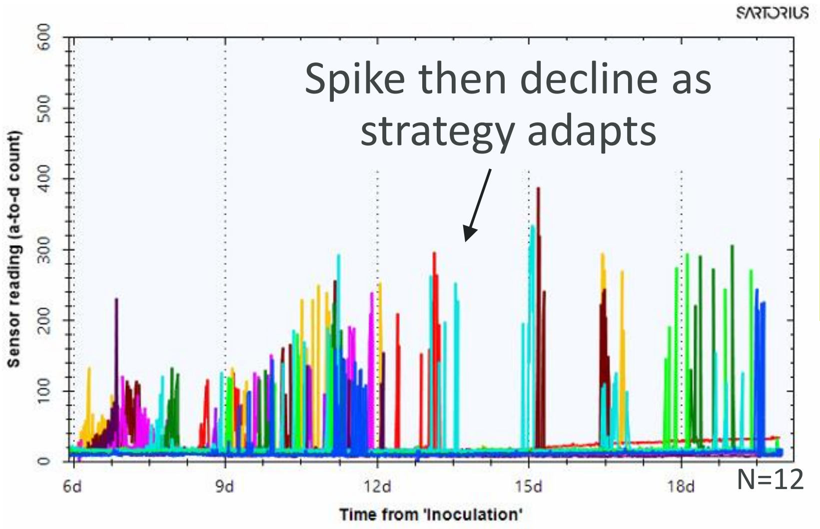
Emergency antifoam based on vessel pressure



Automated foam control

Regular antifoam additions that ramp in frequency and volume based on gas flow rate

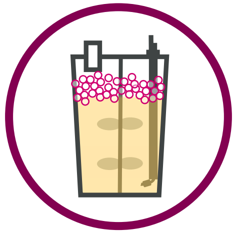
Emergency antifoam based on foam sensor



Foam sensor detects around height of 280 mL

Example Experiments*	Vessels that saw foam outs >10% of volume
Previous foam control strategy (N=20)	30%
Automated Foam Control (N=36)	3%

The Automated Foam Control strategy reduced total foam out volume per experiment by 90% on average



*this table is not comprehensive but showcases several representative experiments

User-guided level control strategy produces consistent working volume

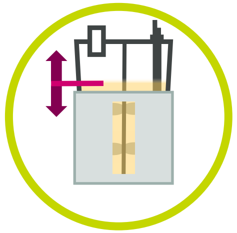
- Four separate inlets and two separate outlets affecting the volume, as well as pipette additions and sampling
- Volume totalizer calculation accurate to ~1% total volume exchange (1-1.5 VVD)

Long-term

- Camera integration
- Built-in level sensor or scale

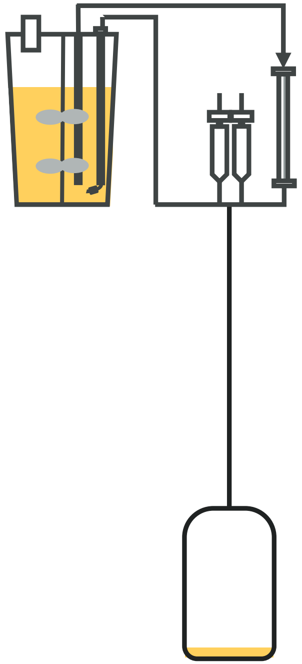
Short-term Fix

- Working volume increased to sit above heating jacket
- Marked ruler tape
- Minimal effort
- Quick daily check by user
- Maintain volume ± 5 mL

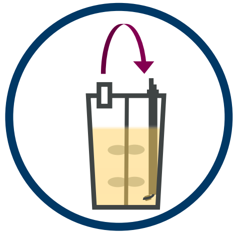


Fully automated seed-to-production transition increases experiment capacity

- Automatically or manually triggered
- Drain vessels stations using bleed function
- Volume calculated based on ratio or target density
- Bioreactors refilled using LH or pump



capacity increase (%) per experiment	
Single clone optimization	33
Clone selection	100



ambr250ht perfusion platform considerably improved by addressing challenges

Automated foam control improves consistency and user experience

- Minimized foam outs
- Reduced frequency of consumable swaps
- Very occasional user interference

User-led level control strategy maintains working volume

- Working volume now maintained within 5 mL

Automated drain-and-fill reduces FTE and risk of contamination

- Full system available to use as production bioreactors
- No additional effort required from scientist

On average, scientist time reduced by at least 50%;
ambr250 ready for scale-down modeling evaluation

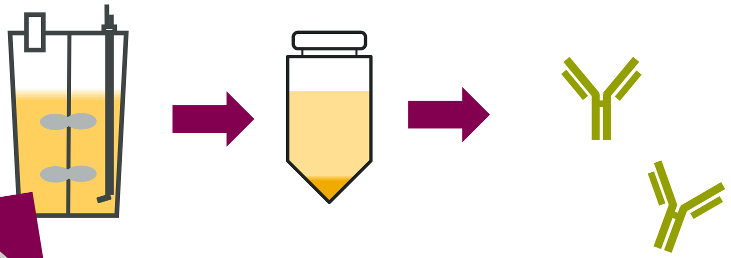
Ambr250 scale-down model evaluation compared to 3L

Transforming scaling metrics for fed-batch culture into perfusion equivalents

Fed-batch:

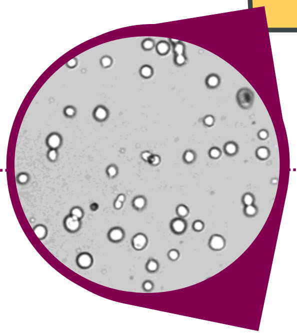
Peak VCD

End-of-run (EOR) viability



Harvest Titer

Product Quality

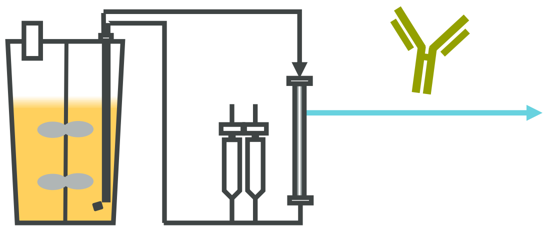


Perfusion:

Early growth rate

Mid-run growth rate

End-of-run (EOR) viability



Average productivity per cell

Total productivity

Product Quality

Multi-project dataset compiled for statistical analysis

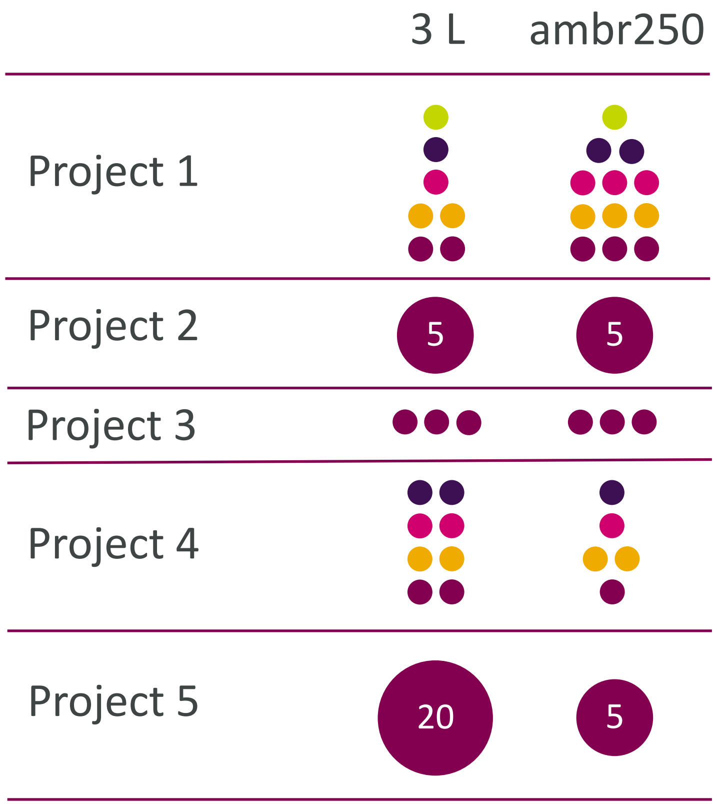
Opportunistic gathering of comparison data between 3-L (representative of larger scales) and ambr250 experiments

Control condition data:

- Easy and difficult to express antibodies produced by CHO host cell lines
- 5 different projects
- 12 separate clones across projects
- Discrepancies caused by equipment failures or other issues were investigated and data removed if appropriate

Details of statistical analysis:

- Analysis performed in jmp
- Tested for linear model effects
- project/clone pairing included as random nested model effect
- Scale included as main model effect
- Controlled for instrumentation differences in cell counters using cross-samples and conversion factors

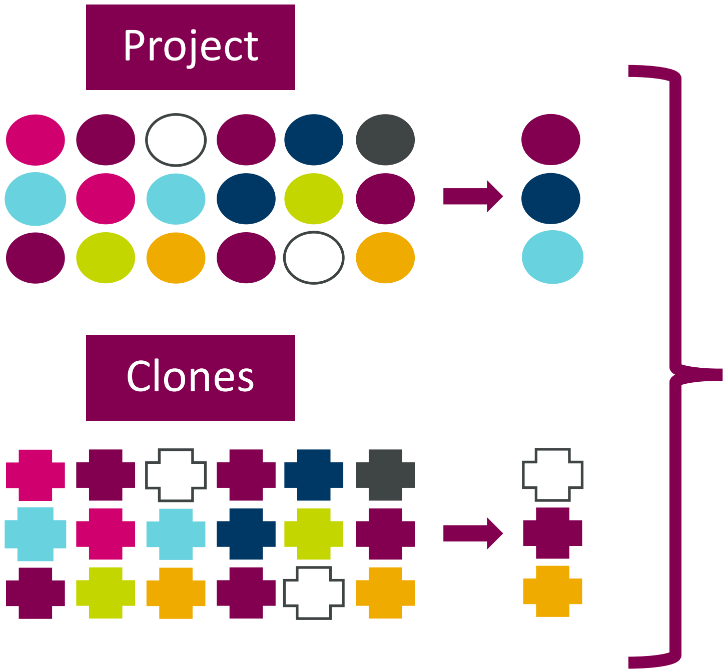


Linear to Mixed Model: Random Nested Variables

A random effects model is used when studying a subset of a larger population

A nested model is used when one set of available factor values are dependent on a primary factor value

Larger Populations



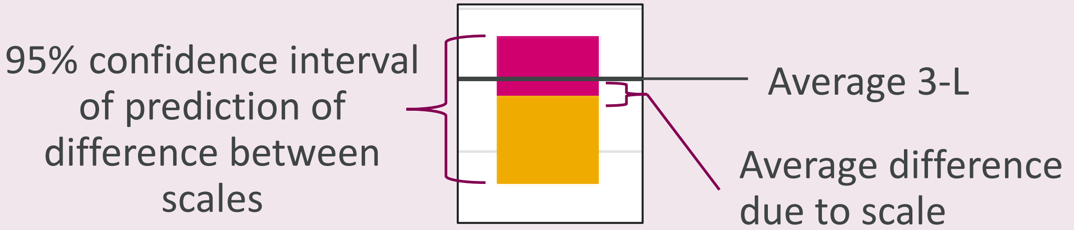
Data Set

Project	Clone	Scale	Data
		3L	…
		Ambr	…
		3L	…
		Ambr	…
⋮	⋮	⋮	

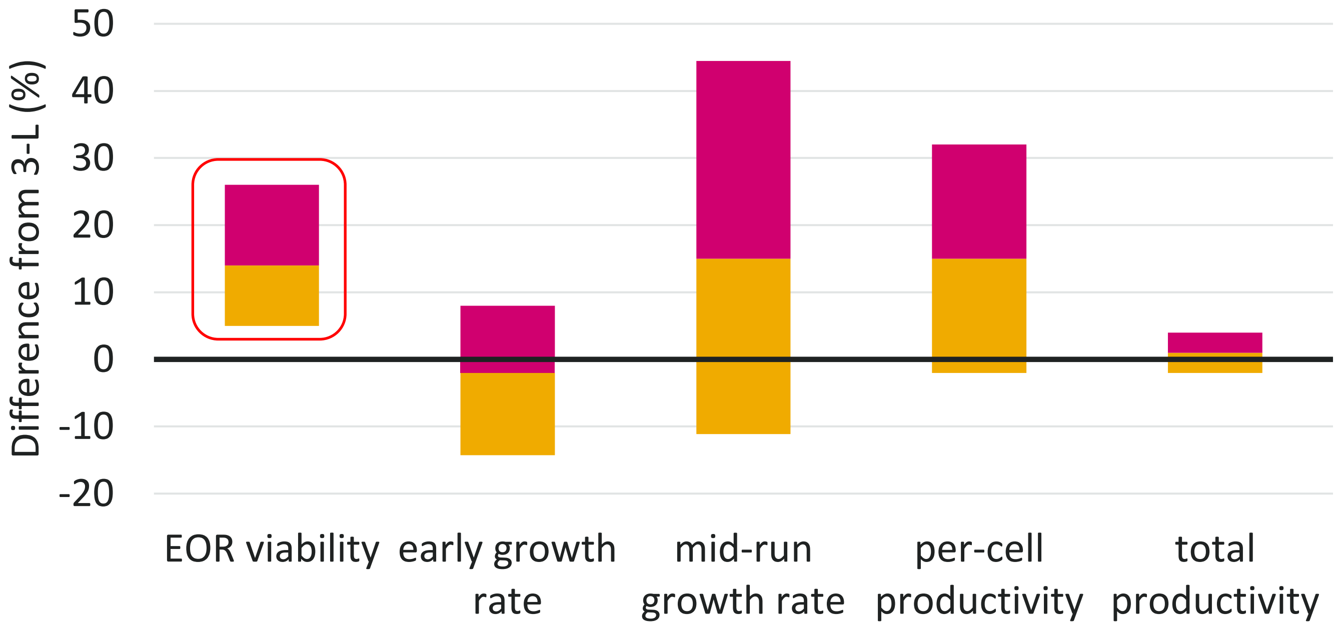
Prediction

Random nested model improves predictions on how future projects and clones not found in the data set will behave

Scale as a factor for cell growth and productivity metrics



Ambr250 scale is not a significant factor if 95% confidence interval overlaps with average 3-L

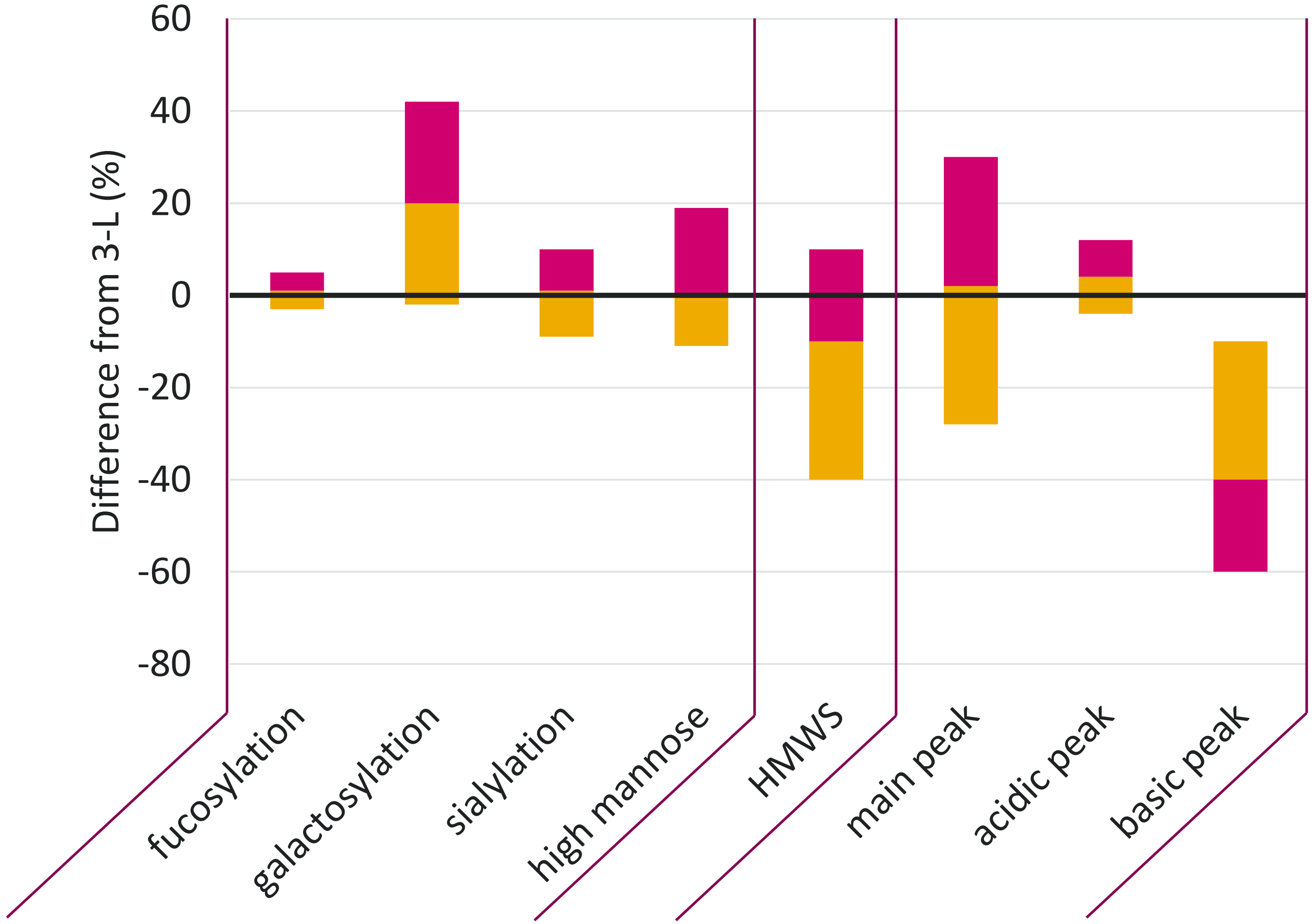


- Scale found to be a significant factor for end-of-run viability
- Scale not found to be a significant factor for early growth rate, md-run growth rate, per-cell productivity, or total productivity

Data from jmp transformed into percentages based on average 3-L value

Scale as a factor for product quality attributes

For all Glycosylation:	N of experiments	
	3L	ambr250
Project 1	4	8
Project 2	2	2
Project 3	2	2
Project 4	2	2
Project 5	11	5
Total	**21**	**19**



- Scale found to be a significant factor for basic peak
- Scale not found to be a significant factor for other product quality attributes

Conclusions

Conclusions

- The ambr250 system offers expanded bioreactor capacity while reducing lab work
- Several important differences between the ambr250 and larger bioreactor scales necessitate scale-down model testing
- Prior to scale down model testing, several challenges related to controlling the working volume, preventing foam outs, and automating drain-and-fill were addressed
- Scale down modeling performed in statistical software found that scale was a significant factor for EOR viability and basic peaks (a product quality attribute)
- Although significant, this level of scale-dependent impact is acceptable and allows the ambr250 to be used as a scale-down model

Acknowledgements

AstraZeneca

Data Science and Modeling

John Murphy

John Patrick Mpindi

Chris Thompson

Cell Culture and Fermentation Sciences

Jennifer Au

George Chacko

Jan Kemper

Nikita Tatwadhia

BioProcess Technologies and Engineering

Alex Gadberry

Anita Kundu

Ken Lee

Samuel Preza

Andrea Squeri

Lorraine Peters

Ashna Dhingra

Christine Ferng

SARTORIUS

Greg Laslo

Ricky Chan

Andrew Tait

Damon Clarke

Don Traul